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Q1. String Character Frequency Counter

CODE

```
s = input().lower()
freq = {}
for char in s:
    if char ≠ "":
        freq[char] = freq.get(char,0) + 1
for char in sorted(freq.keys()):
    print(f"{char} - {freq[char]}")
```

INPUT

```
Hello World
```

OUTPUT

```
  - 1
d - 1
e - 1
h - 1
l - 3
o - 2
r - 1
w - 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()

vowels_chars = "aeiou"
vow_counts = {}
con_counts ={}
total_vow = 0
total_con = 0
for char in s:
    if char.isalpha():
        if char in vowels_chars:
            vow_counts[char] = vow_counts.get(char,0) + 1
            total_vow += 1
        else:
            con_counts[char] = con_counts.get(char,0) + 1
            total_con += 1
if vow_counts:
    max_vow_val = max(vow_counts.values())
    best_vow = sorted([k for k,v in vow_counts.items() if v == max_vow_val])[0]
    most_frequent_vowel = f"{best_vow} ({max_vow_val})"
else:
    most_frequent_vowel = "None (0)"
if con_counts:
    min_con_val = min(con_counts.values())
    best_con = sorted([k for k, v in con_counts.items() if v == min_con_val])[0]
    least_frequent_consonant = f"{best_con} ({min_con_val})"
else:
    least_ferquent_consonant = "None (0)"
if total_con > 0:
    ratio = total_vow/total_con
    ratio_str = f"{ratio:.2f}"
else:
    ratio_str = "0.00"
print(f"Vowels: {total_vow} | Consonants: {total_con} | Most frequent vowel: {most_frequent_vowel} |Least frequent consonant: {least_
frequent_consonant} | Vowel-to-Consonant Ratio: {ratio_str}")
```

INPUT

```
happy python coder
```

OUTPUT

```
Vowels: 4 | Consonants: 12 | Most frequent vowel: o (2) |Least frequent consonant: c (1) | Vowel-to-Consonant Ratio: 0.33
```

Q3. String Compression and Run-Length Encoding

CODE

```
s = input()
res = ""
i = 0
while i < len(s):
    c = 1
    while i + 1 < len(s) and s[i] == s[i+1]:
        c += 1
        i += 1
    res += s[i] + (str(c) if c > 1 else "")
    i += 1
if len(res) ≥ len(s):
    print(f"For input '{s}': Compression not beneficial")
    print(f"Original: {s}")
else:
    print("Compressed:", res)
    dec = ""
    i = 0
    while i < len(res):
        ch = res[i]
        i += 1
        num = ""
        while i < len(res) and res[i].isdigit():
            num += res[i]
        dec += ch * (int(num) if num else 1)
    print("Decompressed:", dec)
```

INPUT

```
aabbbccddddee
```

OUTPUT

```
(no output)
```

Q4. Advanced Text Statistics Analyser

CODE

```
s = input()
c, i = "",0
while i < len(s):
    j = i
    while j < len(s) and s[j] == s[i]:
        j += i
    c += s[i] + s[j] == s[i]:
    i = j
if len(c) ≥ len(s):
    print(f"For input '{s}': Compression not beneficial.")
    print(f"Original: {s}")
else:
    print("Compressed:", c)
    d, k = "", 0
    while k < len(c):
        ch =
```

OUTPUT

```
(no output)
```